

mos_simple_2th: simplified power MOSFET loss model with separate thermal ports

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Abstract

This note describes the Simscape language component `mos_simple_2th`, a simplified power MOSFET model for loss estimation and nominal-value thermal dimensioning. It is the MOSFET counterpart of `igbt_simple_2th`: same port interface, same single-point switching energies scaled with current and voltage, same continuous sample-and-hold states and the same energy-to-power conversion towards two thermal ports, H_t for the channel and H_d for the antiparallel diode. The differences are those imposed by the device: a bidirectional resistive channel with third-quadrant conduction, a body diode that carries the current only during the dead time in synchronous rectification, zero-voltage turn-on, and a diode that is physically on the same chip as the channel.

Revision history

Rev.	Date	Description
00	2026-09-19	First issue.

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1 Scope

The component computes, in a circuit simulation with ideal-switching semiconductors, the conduction and switching losses of a power MOSFET and of its antiparallel diode and delivers them, separately, to two thermal conserving ports. The switching transients are not resolved: the model is not meant for snubber, overvoltage or EMI studies (for those the switching-dynamics model `mos_dyn` exists). The purpose is a nominal-value dimensioning: average and cyclic losses, junction temperature ripple over the output fundamental, heatsink verification.

The modelling choices are the same as in `igbt_simple_2th`, documented in `igbt_simple_doc`; this note repeats only what is needed to read the source and dwells on what is different for a MOSFET. A user of the IGBT block can replace it with the MOSFET block without changing the surrounding model.

2 Interface

Table 1: Ports, inputs and outputs.

Name	Type	Meaning
D, S	electrical	Drain and source. Positive current i from D to S, terminal voltage $v = v_D - v_S$.
Ht	thermal	Channel chip: conduction $v i_t$ plus E_{on} , E_{off} .
Hd	thermal	Diode chip: conduction $v i_d$ plus E_{rr} .
G	input, V	Gate signal. Channel on for $g > V_{th}$; switching events at $V_{th} \pm V_{hyst}$.
Pt, Pd	output, W	Instantaneous losses delivered to H_t and H_d (conduction plus averaged switching power).
Ptflt, Pdflt	output, W	The same, filtered with τ_o for display; not used by the thermal ports.
it, id	output, A	Channel and diode currents. In blocking they carry the leakage $v G_{off}$.

3 Conduction model

3.1 Channel

The channel is a linear resistance, on for $g > V_{th}$ and bidirectional:

$$i_t = \begin{cases} v/R_{on}(T_j) & g > V_{th} \\ v G_{off} & \text{otherwise,} \end{cases} \quad R_{on}(T_j) = R_{DS(on)} [1 + \alpha_R (T_j - T_{ref})], \quad (1)$$

with T_j read from the H_t port. With the default $\alpha_R = 0$ the resistance is the fixed 125 °C value, as in the IGBT model, and the electrical and thermal networks are decoupled. A non-zero α_R closes the electro-thermal loop, which for a MOSFET is worth doing because $R_{DS(on)}$ varies by a factor 2 to 2.5 between 25 °C and 150 °C in silicon and by 1.4 to 1.7 in SiC (Section 8).

Table 2: Parameters. Defaults are placeholders for a generic 1200 V/300 A SiC MOSFET half-bridge at $T_{vj} = 125^\circ\text{C}$ and must be replaced with datasheet values.

Name	Symbol	Default	Meaning
<i>Conduction</i>			
Rds_on	$R_{DS(on)}$	5 m Ω	Channel on-state resistance at T_{ref} .
alpha_R	α_R	0/K	Linear temperature coefficient of $R_{DS(on)}$; 0 = fixed value at T_{ref} .
T_ref	T_{ref}	398.15 K	Reference junction temperature of $R_{DS(on)}$ (125 °C).
Vf	V_f	2.6 V	Diode threshold voltage.
Rd	R_d	5 m Ω	Diode differential resistance.
<i>Gate</i>			
Vth	V_{th}	4 V	Gate-source threshold.
Vhyst	V_{hyst}	1 V	Half hysteresis band of the switching events.
<i>Switching energies, single point</i>			
Eon_ref, Eoff_ref	E_{on}^*, E_{off}^*	12 mJ, 6 mJ	Channel energies at I_{ref}, V_{ref} .
Err_ref	E_{rr}^*	1 mJ	Diode recovery energy at I_{ref}, V_{ref} .
Ic_ref, Vdc_ref	I_{ref}, V_{ref}	300 A, 600 V	Reference conditions of the energies.
k_i, k_v	k_i, k_v	1, 1	Current and voltage exponents of E_{on}, E_{off} .
k_id, k_vd	k_{id}, k_{vd}	1, 1	Current and voltage exponents of E_{rr} .
<i>Numerical</i>			
V_rr_th	$V_{rr,th}$	50 V	Voltage level detecting the recovery event.
i_eps	i_ε	10 mA	Reverse-conduction detection level of i_{dh} .
Goff	G_{off}	$1 \times 10^{-6}/\Omega$	Off-state conductance.
tau_s	τ_s	1 μ s	Sample-and-hold time constant.
tau_p	τ_p	200 μ s	Switching-energy averaging time constant.
tau_o	τ_o	1 ms	Display filter of Ptflt, Pdflt.

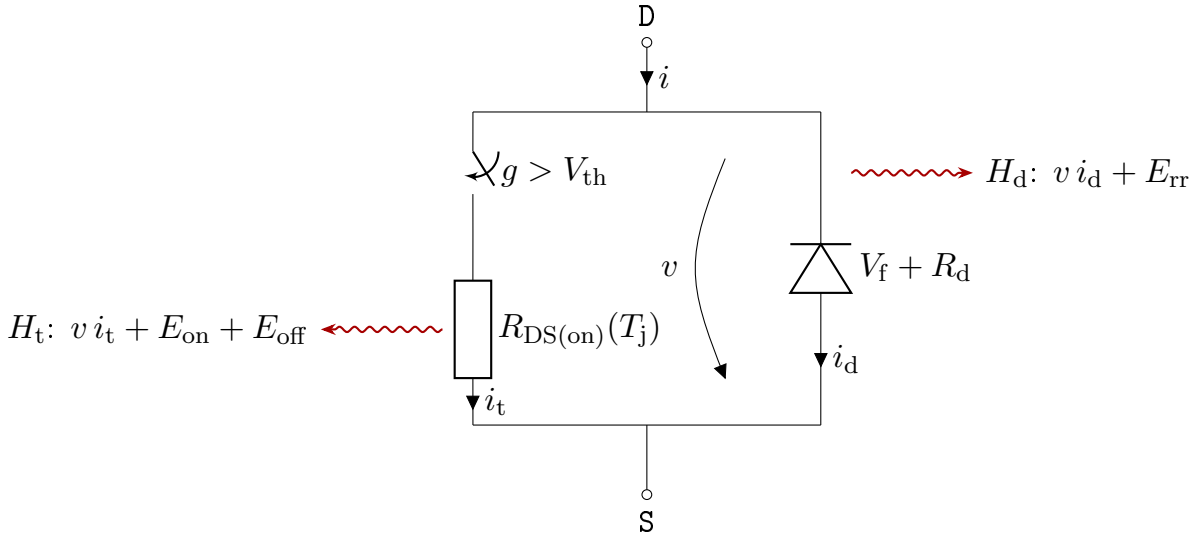


Figure 1: Equivalent circuit of the component. The channel is a bidirectional resistance in series with an ideal switch; the diode conducts for $v < -V_f$. The two branches dissipate towards two distinct thermal ports.

3.2 Diode

The diode is a threshold voltage plus a differential resistance:

$$i_d = \begin{cases} (v + V_f)/R_d & v < -V_f \\ v G_{\text{off}} & \text{otherwise.} \end{cases} \quad (2)$$

Unlike the IGBT model, where R_d was a numerical conditioning term, here both V_f and R_d are physical: the body diode of a SiC MOSFET has a threshold around 2.5 V to 3 V and a slope resistance comparable to $R_{\text{DS(on)}}$, and the two together determine how the current splits during the dead time and how much the diode dissipates.

3.3 Third-quadrant operation

The two branches are in parallel and the channel resistance has no preferential direction, so third-quadrant conduction requires no additional logic. With $i < 0$ and the gate on, $v = i R_{\text{on}}$ is a few hundred millivolts and the diode is off: the current flows in the channel and the loss is charged to H_t . With the gate off the current flows in the diode at $v = -V_f + i R_d$ and the loss is charged to H_d . The diode is therefore active only during the dead times and whenever $|i| R_{\text{on}} > V_f$, when it shares the current with the channel exactly as in the real device.

4 Switching losses

4.1 Scaling law

Each energy is a single datasheet point scaled with the commutated current I and the blocking voltage V :

$$E = E^* \left(\frac{I}{I_{\text{ref}}} \right)^k \left(\frac{V}{V_{\text{ref}}} \right)^{k_v}. \quad (3)$$

The exponents are parameters, with $k = k_v = 1$ by default so that the block reproduces the bilinear law of the IGBT model. For a MOSFET a sub-linear current exponent is often the better fit, in particular for E_{off} , which at low current is dominated by the output-capacitance energy E_{oss} and does not go to zero with I (Section 8). The quantities I and V are not taken at the switching instant, where the solver already sees the post-transition values, but from the held states of Section 4.3.

4.2 Attribution to the two chips

E_{on} and E_{off} are dissipated in the channel of the device that switches, and are added to the accumulator E_t at the rising and falling edges of its own gate. E_{rr} is dissipated in the diode of *this* device when the *complementary* switch of the leg turns on. That instant is not visible from the gate; it is detected locally, as in the IGBT model, as the rising edge of the terminal voltage above $V_{\text{rr,th}}$, and the energy is added to the accumulator E_d .

E_{on} and E_{rr} are not the same energy counted twice. When the channel turns on against a conducting diode, the recovery current of that diode flows through the channel at full voltage and its loss is part of the datasheet E_{on} of the switching device; the loss inside the recovering diode is E_{rr} . Both are used, provided the datasheet E_{on} was measured with the same diode that freewheels in the application (Section 8).

4.3 Sample-and-hold states

Three continuous states with time constant τ_s store the pre-commutation quantities. The channel current, used for E_{on} and E_{off} :

$$\tau_s \dot{i}_h = \begin{cases} i_t - i_h & g > V_{\text{th}} & (\text{track, channel on}) \\ -i_h & v < -V_f & (\text{flush, diode conducting}) \\ 0 & \text{otherwise} & (\text{hold, blocking}) \end{cases} \quad (4)$$

At turn-off i_h is the current being switched off. At turn-on it is the current switched off one period earlier if the device has been blocking since, which is the current about to be switched on; it is zero if the diode of the device has conducted in the meantime, because the channel then closes at zero voltage. The flush branch is the MOSFET-specific addition: in the IGBT model i_h only tracked and held, and a turn-on with the own diode conducting was charged E_{on} at the previous turn-off current.

The blocking voltage, used by all three energies:

$$\tau_s \dot{v}_h = \begin{cases} v - v_h & g < V_{\text{th}} \wedge v > 0 \\ 0 & \text{otherwise.} \end{cases} \quad (5)$$

The reverse-current magnitude, used for E_{rr} :

$$\tau_s \dot{i}_{\text{dh}} = \begin{cases} -i - i_{\text{dh}} & i < -i_\epsilon & (\text{track, reverse conduction}) \\ -i_{\text{dh}} & g > V_{\text{th}} \wedge i > i_\epsilon & (\text{flush, forward channel}) \\ 0 & \text{otherwise} & (\text{hold, blocking}) \end{cases} \quad (6)$$

This is the second MOSFET-specific change. The IGBT version tracked the *diode* current, $v < -V_f$, and flushed whenever the transistor was gated on. In a MOSFET with synchronous rectification the reverse current is in the channel for most of the period and in the diode only during the dead time before the recovery, typically 0.2 μs to 1 μs for SiC, too short for a tracker with $\tau_s \approx 1 \mu\text{s}$ to settle. The state therefore tracks the total reverse current, whichever path it takes: what is recovered is the current that the diode carries at the end of the dead time, and that is the load current regardless of how it was shared a moment before. The flush condition is restricted to *forward* channel conduction, so that gating the channel on in the third quadrant does not erase the held value.

4.4 Events

The three events are a single **when/elsewhen** chain, because two separate **when** clauses updating the same event variable are rejected by the compiler, and the branches are mutually exclusive by construction. With $[x]^+ = (x + |x|)/2 = \max(x, 0)$:

$$g \uparrow (V_{\text{th}} + V_{\text{hyst}}) : E_t \leftarrow E_t + E_{\text{on}}^* \left(\frac{[i_h]^+}{I_{\text{ref}}} \right)^{k_i} \left(\frac{|v_h|}{V_{\text{ref}}} \right)^{k_v} \quad (7)$$

$$g \downarrow (V_{\text{th}} - V_{\text{hyst}}) : E_t \leftarrow E_t + E_{\text{off}}^* \left(\frac{[i_h]^+}{I_{\text{ref}}} \right)^{k_i} \left(\frac{|v_h|}{V_{\text{ref}}} \right)^{k_v} \quad (8)$$

$$v \uparrow V_{\text{rr,th}} : E_d \leftarrow E_d + E_{\text{rr}}^* \left(\frac{|i_{\text{dh}}|}{I_{\text{ref}}} \right)^{k_{id}} \left(\frac{|v_h|}{V_{\text{ref}}} \right)^{k_{vd}}. \quad (9)$$

The positive part $[i_h]^+$ replaces the $|i_h|$ of the IGBT model: a channel that is switched with negative current commutates with its own diode, at zero voltage, and neither E_{on} nor E_{off} is generated. At the turn-off of a device carrying forward current the gate edge and the voltage edge coincide; the chain executes only the first branch, and the E_{rr} branch is skipped, which is correct since i_{dh} has been flushed anyway.

4.5 Behaviour over the two half periods

Figure 2 follows one switching period of the device in the half period of the output fundamental in which it carries reverse current (synchronous rectification). Table 3 summarises both half periods.

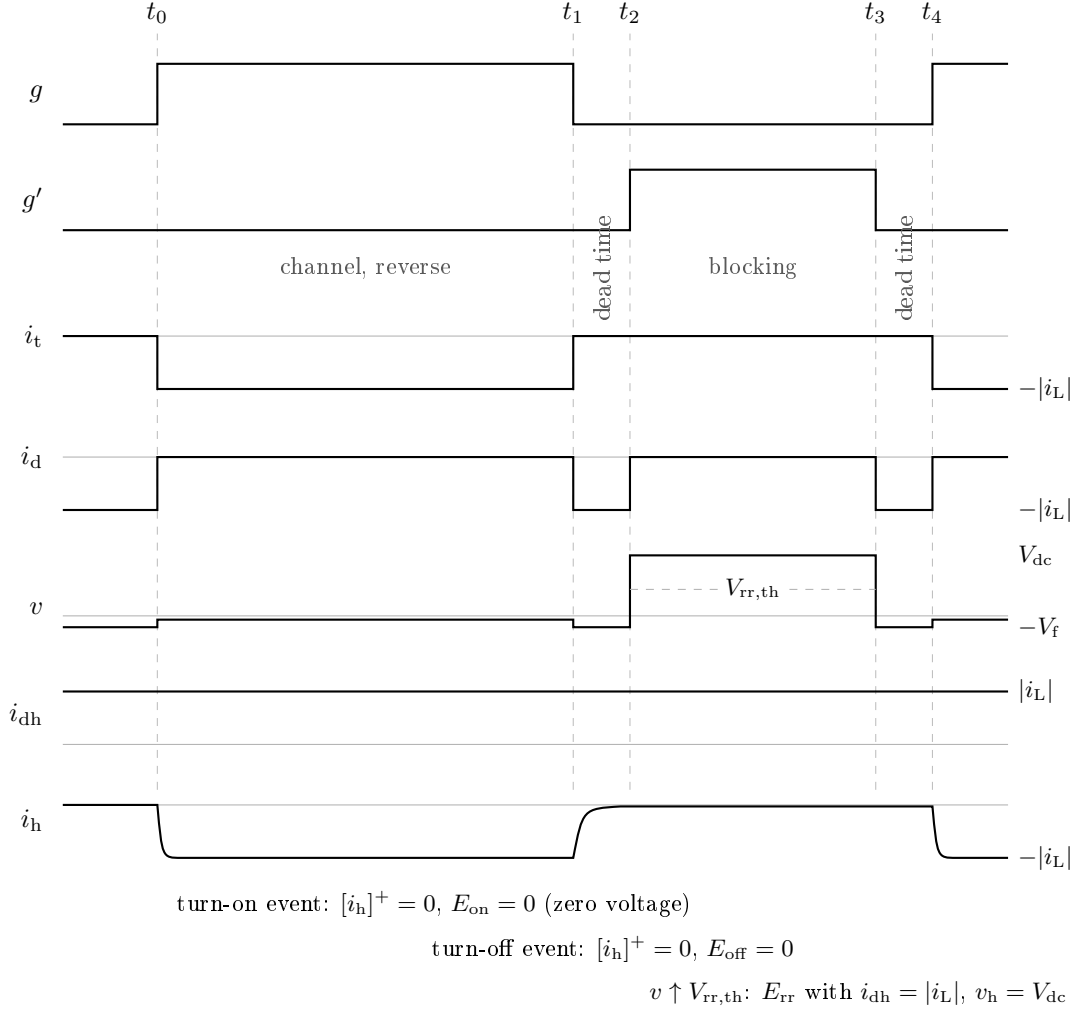


Figure 2: One switching period in the half period with reverse load current. t_0 : channel on, current moves from the diode to the channel. t_1 : channel off, current back to the diode. t_2 : complementary switch on, recovery, v rises to V_{dc} . t_3 : complementary switch off, dead time. t_4 : next turn-on at zero voltage. i_{dh} tracks $|i|$ through channel and diode alike and holds during blocking; i_h is negative or flushed, so no channel energy is generated.

At the zero crossings of the load current the sign of i changes inside a switching period and one event of each kind is evaluated with a partially tracked or partially flushed state. The error is one switching event per fundamental half period and is irrelevant for a thermal dimensioning.

5 Energy-to-power conversion and thermal ports

The event variables E_t and E_d are staircases. Each is converted to a power by a first-order lag with time constant τ_p :

$$\tau_p \dot{E}_{lp} = E - E_{lp}, \quad P_{sw} = \frac{E - E_{lp}}{\tau_p} = \dot{E}_{lp}. \quad (10)$$

Since P_{sw} is the derivative of E_{lp} and $E_{lp} \rightarrow E$ after the last event, the energy delivered to the thermal port equals the accumulated energy: the lag only spreads each step over a few τ_p , without

Table 3: Held states and generated energies in the two half periods of the output fundamental.

	Forward current ($i > 0$)	Reverse current ($i < 0$)
Channel on	carries i ; i_h tracks $i > 0$; i_{dh} flushed	carries i ; i_h tracks $i < 0$; i_{dh} tracks $ i $
Own turn-off	$E_{off}(i_h > 0, v_h)$; v rises, E_{rr} branch skipped (chain)	$E_{off} = 0$; current to own diode
Dead time after own turn-off	device blocks, v_h tracks V_{dc} ; i_h , i_{dh} hold	own diode conducts; i_{dh} tracks, i_h flushes
Complementary turn-on	no edge on this device	$v \uparrow V_{rr,th}$: $E_{rr}(i_{dh} = i , v_h)$
Complementary on	blocking; complementary diode inactive	blocking; v_h tracks V_{dc} ; i_{dh} holds $ i $
Dead time before own turn-on	complementary diode conducts; device blocks; i_h holds $i > 0$	own diode conducts; i_{dh} tracks
Own turn-on	$E_{on}(i_h > 0, v_h)$: hard switching	$E_{on} = 0$: $i_h \leq 0$, zero-voltage turn-on

changing its integral. The total losses are

$$P_t = v i_t + \frac{E_t - E_{lp,t}}{\tau_p}, \quad P_d = v i_d + \frac{E_d - E_{lp,d}}{\tau_p}, \quad (11)$$

and are bound to the thermal ports with the sign convention of the thermal domain, positive Q flowing into the component: $Q_t = -P_t$ and $Q_d = -P_d$. τ_p must be a few switching periods, so that the thermal network sees a smooth power, and much shorter than the smallest thermal time constant of interest.

5.1 Two ports on one chip

In an IGBT module the transistor and the freewheeling diode are two chips with two thermal impedances, and the two ports give two junction temperatures. In a MOSFET the body diode is the drain-body junction of the same die: channel and diode losses heat the same silicon. In that case the two ports must be connected to the *same* junction node, feeding a single $Z_{th(j-c)}$ (Figure 3); the split between H_t and H_d remains available through P_t and P_d as a loss breakdown and for the cross-check against the vendor’s loss calculator, but it is not two temperatures. The ports are kept separate for the modules that do have a discrete antiparallel diode chip (SiC MOSFET with SiC Schottky, silicon MOSFET with external fast diode); there V_f , R_d and E_{rr}^* describe the external chip, whose lower threshold keeps the body diode off, and H_d gets its own Z_{th} .

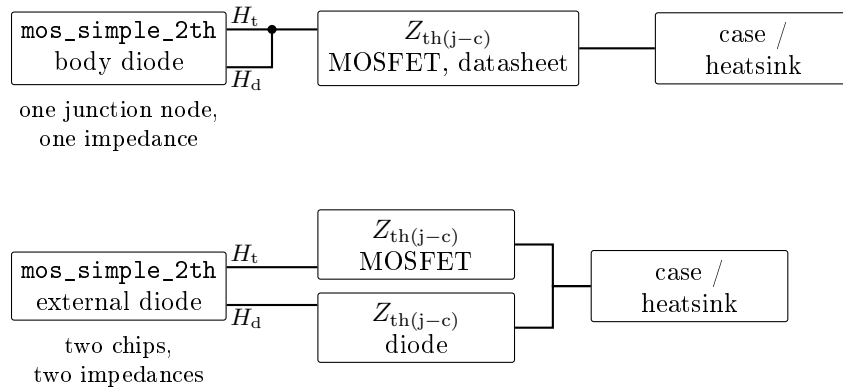


Figure 3: Connection of the thermal ports. Top: intrinsic body diode, one junction. Bottom: discrete antiparallel diode chip, two junctions.

6 Outputs

P_t and P_d are the powers actually delivered to the thermal ports. P_{t_flt} and P_{d_flt} are the same signals through a further lag τ_o , for scopes and for the average-loss readout; they do not enter the thermal network, so τ_o can be raised for readability without altering the junction temperature. The current outputs i_t , i_d carry the leakage $v G_{off}$ while blocking, of the order of a milliampere at 1 kV; if they are integrated for a charge balance this offset must be subtracted.

7 Reference circuit

The parameters and the timing of Figure 2 refer to the half-bridge leg of Figure 4, which is also the datasheet test circuit for the switching energies. The load current i_L is positive out of the switch node; with $i_L > 0$ the upper device works in the forward-current column of Table 3 and the lower one in the reverse-current column.

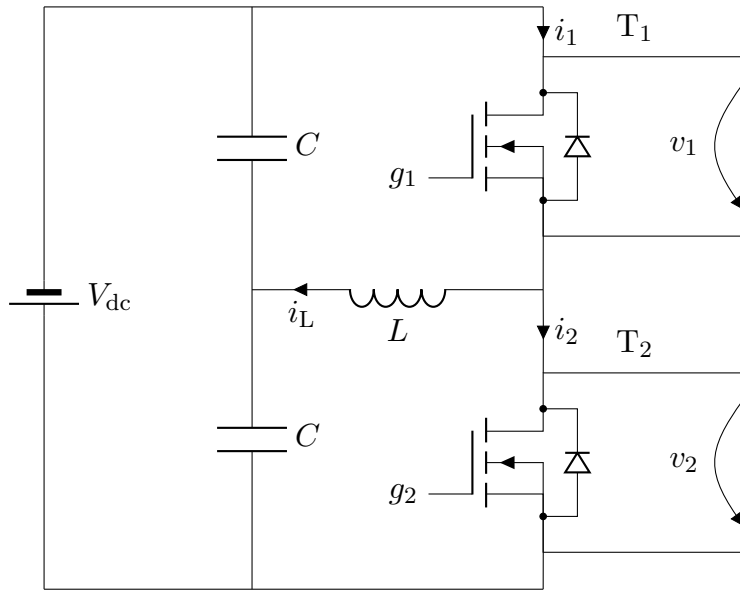


Figure 4: Half-bridge reference leg. Each device is one instance of the component, gate signals g_1 , g_2 complementary with dead time.

8 Parameterisation from the datasheet

Channel. $R_{DS(on)}$ is the typical value at the gate voltage of the application and at T_{ref} , read from the $R_{DS(on)}(T_j)$ curve, not the 25 °C headline value. If the electro-thermal loop is closed, α_R is the secant slope between two datasheet temperatures around the operating point, e.g. between 100 °C and 150 °C, normalised to the value at T_{ref} ; typical values around 125 °C are $5 \times 10^{-3}/K$ to $7 \times 10^{-3}/K$ for silicon and $2.5 \times 10^{-3}/K$ to $3.5 \times 10^{-3}/K$ for SiC. The linear law is adequate within about $\pm 40 K$ of T_{ref} .

Diode. V_f and R_d are the intercept and slope of a straight line fitted to the body-diode characteristic at T_{ref} , with the gate at its *off* voltage, between roughly $0.3 I_{nom}$ and I_{nom} . For SiC the characteristic depends on the negative gate bias; use the value of the application. With a discrete antiparallel diode, fit that diode instead.

Switching energies. E_{on}^* , E_{off}^* are read at I_{ref} , V_{ref} , T_{ref} and at the external gate resistance of the application; the dependence on R_G is strong for SiC and is not modelled. The test circuit

matters more than for an IGBT: most MOSFET datasheets measure E_{on} in a half-bridge with the body diode of the complementary device as freewheeling diode, some with an external SiC Schottky. Only the first is consistent with a leg of two identical MOSFETs; the second underestimates E_{on} by the recovery-induced term. E_{rr}^* is the body-diode recovery energy, often listed as E_{rec} or given through Q_{rr} , in which case $E_{\text{rr}}^* \approx V_{\text{ref}} Q_{\text{rr}}$ is the usual estimate; for a SiC body diode it is small and mostly capacitive.

Exponents. With $k_i = k_v = 1$ the block behaves as `igbt_simple_2th`. If the datasheet gives $E(I)$ curves, a fit over the operating range typically returns $k_i \approx 1$ for E_{on} , $k_i \approx 0.6$ to 0.9 for E_{off} and, for the voltage, $k_v \approx 1.2$ to 1.5 for E_{on} and ≈ 1.0 to 1.3 for E_{off} . Since the model has one pair of exponents for both channel energies, choose them on the energy that dominates at the operating point, or split the difference. For a SiC body diode $k_{id} \approx 0.3$ to 0.5 reflects the weak current dependence of a mainly capacitive recovery.

Numerical parameters. τ_s must be shorter than about one third of the shortest on-time and of the dead time, otherwise the trackers do not settle; $1\text{ }\mu\text{s}$ suits IGBT-like dead times, SiC legs with 200 ns to 500 ns dead time need $0.1\text{ }\mu\text{s}$ to $0.2\text{ }\mu\text{s}$. $V_{\text{rr,th}}$ must exceed the largest on-state voltage of both branches and stay well below the minimum dc-link voltage; 50 V is adequate for any device above 600 V . τ_p between two and five switching periods. i_e must exceed the leakage $V_{\text{dc}} G_{\text{off}}$.

Thermal network. $Z_{\text{th(j-c)}}$ of the MOSFET in Foster form from the datasheet, connected as in Figure 3; then $R_{\text{th(c-s)}}$ per module and the heatsink. For a body-diode device the datasheet may list a separate Z_{th} for the diode: it refers to the same chip in diode operation and is not a second path; use the MOSFET one and tie the ports.

9 Validity limits

- Ideal switching: no dv/dt , di/dt , overshoot or recovery current. Use `mos_dyn` for those.
- Single-point energies with power-law scaling; no dependence on R_G and, for the energies, on temperature. All parameters are meant at 125°C .
- $R_{\text{DS(on)}}$ temperature dependence linear and optional; diode characteristic fixed.
- The dead-time loss of the body diode is modelled through its conduction drop; the additional loss of a SiC body diode due to bipolar degradation is not.
- E_{on} at the current zero crossings of the load uses a partially settled i_h : one switching event per fundamental half period is misestimated.
- Short on-times, below $3\tau_s$, leave i_{dh} partially flushed and generate a small spurious E_{rr} at the device's own turn-off.
- Recovery detection assumes that a rising edge of v above $V_{\text{rr,th}}$ is caused by the complementary switch. In topologies where the device voltage rises for other reasons (resonant transitions, dc-link steps) the detection must be reviewed.
- With the intrinsic body diode the two thermal ports are one junction: see Section 5.1.

10 Differences with respect to `igbt_simple_2th`

A Source listing

Table 4: Summary of the differences between the two components.

	igbt_simple_2th	mos_simple_2th
Transistor conduction	$V_{CE,sat}$ fixed, R_{on} numerical, unidirectional	$R_{DS(on)}(T_j)$ resistive, bidirectional
Diode conduction	V_f fixed, R_d numerical	$V_f + R_d$ both physical
Third quadrant	diode only	channel when gated, diode in dead time
i_h	track / hold	track / flush while own diode conducts / hold
i_{dh}	tracks diode current, flushed while gated on	tracks total reverse current, flushed only in forward channel conduction
Channel energies	$ i_h $	$[i_h]^+$: none in the third quadrant
Scaling	bilinear	power laws, exponents as parameters (default bilinear)
Thermal ports	two chips, two junctions	one junction for body-diode devices, two for discrete diode
Outputs	Pt, Pd (plus user additions)	Pt, Pd, Pt_flt, Pd_flt, it, id

```

1 component mos_simple_2th
2 % MOS_SIMPLE_2TH Simplified power MOSFET + antiparallel diode, separate thermal ports
3 %
4 % Companion of IGBT_SIMPLE_2TH: same interface, same sample-and-hold scheme,
5 % same energy-to-power conversion. Intended for loss estimation and
6 % nominal-value thermal dimensioning, not for switching-transient studies.
7 %
8 % Conduction : channel = bidirectional resistance Rds_on, optional linear
9 %               temperature coefficient alpha_R (default 0 = fixed value at
10 %              T_ref); diode = Vf + Rd in parallel with the channel.
11 %              Third-quadrant conduction follows from the parallel connection:
12 %              current in the channel while the gate is on (synchronous
13 %              rectification), in the diode during the dead time.
14 % Switching   : single-point Eon/Eoff/Err scaled with the commutated current
15 %               and the blocking voltage by power laws (default exponents 1):
16 %               E = E_ref * (I/Ic_ref)^k_i * (V/Vdc_ref)^k_v
17 %               Ht : channel -> conduction v*i_t + Eon + Eoff
18 %               Hd : diode   -> conduction v*i_d + Err
19 %
20 % Thermal ports
21 % -----
22 % If the diode is the intrinsic body diode, channel and diode are the SAME
23 % chip: connect Ht and Hd to the same junction node and one Zth(j-c). The
24 % split is then a loss breakdown, not two temperatures. Keep the ports
25 % separate only for modules with a discrete antiparallel diode chip (e.g.
26 % SiC MOSFET + SiC Schottky); Vf, Rd and Err_ref then describe that chip.
27 %
28 % Sample-and-hold states (continuous, time constant tau_s)
29 % -----
30 % i_h : channel current. Tracked while the gate is on, flushed while the
31 %       diode conducts, held while blocking. At turn-off it is the current
32 %       being switched off; at turn-on it is the current about to be
33 %       switched on if the device was blocking, zero if its diode was
34 %       conducting (zero-voltage turn-on, Eon = 0). Negative values (third
35 %       quadrant) give Eon = Eoff = 0 through the max(i_h, 0) in the events.
36 % v_h : blocking voltage. Tracked while off and forward biased, held
37 %       otherwise.
38 % i_dh : reverse-current magnitude before the recovery. Tracked while the
39 %        device conducts in reverse THROUGH CHANNEL OR DIODE, flushed while
40 %        the channel conducts forward current, held while blocking. Unlike
41 %        the IGBT version it must include the reverse channel current: in
42 %        synchronous rectification the diode carries the current only during

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43 %           the dead time preceding the recovery, too short for a tracker.
44 %
45 % Err detection: rising edge of the terminal voltage above V_rr_th, i.e. the
46 % turn-on of the complementary switch. In the half period in which the device
47 % conducts forward current the same edge occurs at its own turn-off, but i_dh
48 % has been flushed, so Err = 0.
49 %
50 % Datasheet check: Eon must be measured with the same diode that freewheels
51 % in the application (half-bridge test with the body diode). Eon measured
52 % against an external SiC Schottky misses the recovery-induced part of the
53 % turn-on loss. Eon (dissipated in the switching channel) and Err (dissipated
54 % in the recovering diode) are two different energies: both are used.
55 %
56 % Default parameters: generic 1200 V / 300 A SiC MOSFET half-bridge at
57 % Tvj = 125 degC. Placeholders, to be replaced with datasheet values.
58
59 nodes
60     p = foundation.electrical.electrical; % D:left
61     n = foundation.electrical.electrical; % S:right
62     Ht = foundation.thermal.thermal;      % Ht:left
63     Hd = foundation.thermal.thermal;      % Hd:right
64 end
65
66 inputs
67     g = { 0, 'V' };                      % G:left
68 end
69
70 outputs
71     Pt = { 0, 'W' };                      % Pt:right
72     Pd = { 0, 'W' };                      % Pd:right
73     Pt_flt = { 0, 'W' };                  % Pt_flt:right
74     Pd_flt = { 0, 'W' };                  % Pd_flt:right
75     it = { 0, 'A' };                      % it:right
76     id = { 0, 'A' };                      % id:right
77 end
78
79 parameters
80     % --- conduction (Tvj = T_ref, terminal values) ---
81     Rds_on = { 5.0, 'mOhm' }; % Channel on-state resistance at T_ref
82     alpha_R = { 0, '1/K' }; % Rds_on linear temperature coefficient (0 = fixed)
83     T_ref = { 398.15, 'K' }; % Reference junction temperature (125 degC)
84     Vf = { 2.6, 'V' }; % Diode threshold voltage
85     Rd = { 5.0, 'mOhm' }; % Diode differential resistance
86
87     % --- gate ---
88     Vth = { 4, 'V' }; % Gate-source threshold voltage
89     Vhyst = { 1, 'V' }; % Half hysteresis band for the switching events
90
91     % --- switching, single point ---
92     Eon_ref = { 12, 'mJ' }; % Turn-on energy at Ic_ref, Vdc_ref
93     Eoff_ref = { 6, 'mJ' }; % Turn-off energy at Ic_ref, Vdc_ref
94     Err_ref = { 1, 'mJ' }; % Diode recovery energy at Ic_ref, Vdc_ref
95     Ic_ref = { 300, 'A' }; % Reference current of the energies
96     Vdc_ref = { 600, 'V' }; % Reference voltage of the energies
97     k_i = { 1, '1' }; % Current exponent of Eon and Eoff
98     k_v = { 1, '1' }; % Voltage exponent of Eon and Eoff
99     k_id = { 1, '1' }; % Current exponent of Err
100    k_vd = { 1, '1' }; % Voltage exponent of Err
101
102    % --- numerical ---
103    V_rr_th = { 50, 'V' }; % Blocking detection level for the Err event
104    i_eps = { 10, 'mA' }; % Reverse conduction detection level for i_dh
105    Goff = { 1e-6, '1/Ohm' }; % Off-state conductance
106    tau_s = { 1, 'us' }; % Sample&hold time constant (<< on-time and dead time)

```

```

107     tau_p   = { 200,      'us'   }; % Loss averaging time constant (2-5 x Tsw)
108     tau_o   = { 1,       'ms'   }; % Display filter time constant of Pt_flt, Pd_flt
109 end
110
111 variables
112     i       = { 0, 'A' };           % terminal current, D -> S
113     v       = { 0, 'V' };           % terminal voltage, D - S
114     i_t     = { 0, 'A' };           % channel current
115     i_d     = { 0, 'A' };           % diode current
116     Tj      = { 398.15, 'K' };      % channel junction temperature (Ht)
117     i_h     = { 0, 'A' };           % held channel current for Eon/Eoff
118     i_dh    = { 0, 'A' };           % held reverse current before recovery
119     v_h     = { 0, 'V' };           % held blocking voltage
120     E_lpt   = { 0, 'J' };           % low-pass state, channel switching energy
121     E_lpd   = { 0, 'J' };           % low-pass state, diode recovery energy
122     Pt_f    = { 0, 'W' };           % display filter state, channel loss
123     Pd_f    = { 0, 'W' };           % display filter state, diode loss
124     Qt      = { 0, 'W' };           % heat flow into the component through Ht
125     Qd      = { 0, 'W' };           % heat flow into the component through Hd
126 end
127
128 variables (Event = true)
129     E_t     = { 0, 'J' };           % cumulative channel switching energy
130     E_d     = { 0, 'J' };           % cumulative diode recovery energy
131 end
132
133 branches
134     i : p.i -> n.i;
135     Qt : Ht.Q -> *;                 % positive Q = heat flowing INTO the component
136     Qd : Hd.Q -> *;
137 end
138
139 equations
140     assert(Rds_on > 0)
141     assert(alpha_R >= 0)
142     assert(Rd > 0)
143     assert(Vf > 0)
144     assert(Ic_ref > 0)
145     assert(Vdc_ref > 0)
146     assert(Vhyst > 0)
147     assert(tau_s > 0)
148     assert(tau_p > 0)
149     assert(tau_o > 0)
150     assert(i_eps > 0)
151     assert(V_rr_th > Vf)
152
153     v == p.v - n.v;
154     Tj == Ht.T;
155     i == i_t + i_d;
156
157     % ----- conduction -----
158     let
159         Ron = Rds_on*(1 + alpha_R*(Tj - T_ref));
160     in
161         if g > Vth
162             i_t == v/Ron;           % bidirectional channel
163         else
164             i_t == v*Goff;
165         end
166     end
167
168     if v < -Vf
169         i_d == (v + Vf)/Rd;
170     else

```

```

171         i_d == v*Goff;
172     end
173
174     % ----- sample & hold -----
175     % Channel current: tracked while ON, flushed while the diode conducts
176     % (next turn-on is at zero voltage), held while blocking.
177     if g > Vth
178         i_h.der == (i_t - i_h)/tau_s;
179     elseif v < -Vf
180         i_h.der == -i_h/tau_s;
181     else
182         i_h.der == 0;
183     end
184
185     % Blocking voltage: tracked while OFF and forward biased.
186     if (g < Vth) && (v > 0)
187         v_h.der == (v - v_h)/tau_s;
188     else
189         v_h.der == 0;
190     end
191
192     % Reverse current magnitude: tracked while conducting in reverse (channel
193     % or diode), flushed while the channel conducts forward current, held
194     % while blocking.
195     if i < -i_eps
196         i_dh.der == (-i - i_dh)/tau_s;
197     elseif (g > Vth) && (i > i_eps)
198         i_dh.der == -i_dh/tau_s;
199     else
200         i_dh.der == 0;
201     end
202
203     % ----- losses -----
204     E_lpt.der == (E_t - E_lpt)/tau_p;
205     E_lpd.der == (E_d - E_lpd)/tau_p;
206
207     Pt == v*i_t + (E_t - E_lpt)/tau_p;
208     Pd == v*i_d + (E_d - E_lpd)/tau_p;
209
210     Qt == -Pt;
211     Qd == -Pd;
212
213     % ----- outputs -----
214     Pt_f.der == (Pt - Pt_f)/tau_o;
215     Pd_f.der == (Pd - Pd_f)/tau_o;
216     Pt_flt == Pt_f;
217     Pd_flt == Pd_f;
218     it == i_t;
219     id == i_d;
220 end
221
222 events
223     % One when/elsewhen chain: the branches are mutually exclusive by
224     % construction, so no event variable is updated by two event equations.
225     % (i_h + abs(i_h))/2 = max(i_h, 0): no energy for third-quadrant switching.
226
227     % turn-on of the channel
228     when edge(g > (Vth + Vhyst))
229         E_t = E_t + Eon_ref ...
230             * ((i_h + abs(i_h))/(2*Ic_ref))^k_i ...
231             * (abs(v_h)/Vdc_ref)^k_v;
232
233     % turn-off of the channel
234     elsewhen edge(g < (Vth - Vhyst))

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```
235     E_t = E_t + Eoff_ref ...
236         * ((i_h + abs(i_h))/(2*Ic_ref))^k_i ...
237         * (abs(v_h)/Vdc_ref)^k_v;
238
239     % reverse recovery of the diode, forced by the complementary switch
240     elseif edge(v > V_rr_th)
241         E_d = E_d + Err_ref ...
242             * (abs(i_dh)/Ic_ref)^k_id ...
243             * (abs(v_h)/Vdc_ref)^k_vd;
244     end
245 end
246
247 end
```